

Boundary Stratification of the Centered Quartet Manifold

Discriminant Factorization and Uniform Collapse to a Double Pair

Liang Wang

July 2026

Abstract

The finite shape clock of RH-214 moves monotonically in the axial coordinate u , and RH-215 finds a power-gap model to be the best of three finite extrapolants. Neither result proves a limit. This paper instead answers the conditional geometric question exactly: what happens to a centered conjugate quartet if u approaches the boundary value one?

For the canonical shape polynomial $Q_{u,\eta}$, we prove

$$\text{Disc } Q_{u,\eta} = 256(1-u)^2(1-\eta^2)[4u + (1-u)^2\eta^2]^2.$$

Hence degeneracy occurs precisely on the axial edge $u = 1$, on either edge $|\eta| = 1$, or at the isolated point $(u, \eta) = (0, 0)$. The strata correspond respectively to two real double roots, one real double pair, and coincident imaginary conjugate pairs.

Uniformly for $|\eta| \leq 1$, the root multiset approaches $\{1, 1, -1, -1\}$ as $u \rightarrow 1$, with an explicit matching bound

$$d_{\text{root}} \leq \sqrt{2(1-u)} + \frac{1-u}{1+\sqrt{u}}.$$

The coefficient distance to $(z^2 - 1)^2$ is exactly $4(1-u)$ in maximum norm. Thus axial convergence, coefficient convergence, and root-multiset convergence are equivalent. Six hundred random checks verify the formulas to roundoff. The physical data move toward this stratum but reach only $u \approx 0.718$ at the finest anchor; no physical convergence claim is made.

1 Conditional geometry without a convergence claim

RH-213 gives the exact canonical roots

$$a \pm ib, \quad -a \pm id, \tag{1}$$

where

$$a = \sqrt{u}, \quad qqadb^2 = (1-u)(1+\eta), \quad qqadd^2 = (1-u)(1-\eta) \tag{2}$$

for $(u, \eta) \in [0, 1] \times [-1, 1]$ [1]. The polynomial is

$$Q_{u,\eta}(z) = z^4 + (2-4u)z^2 + 4\sqrt{u}(1-u)\eta z + 1 - \eta^2(1-u)^2. \tag{3}$$

RH-214 observes finite monotonicity of u and RH-215 tests three candidate tails [2, 3]. We do not assume any of those tails. All results below hold for an arbitrary point of the compact shape rectangle.

2 Quadratic factorization

Write

$$Q_+(z) = (z - a)^2 + b^2, \text{ and } Q_-(z) = (z + a)^2 + d^2, \text{ and } Q = Q_+ Q_-. \quad (4)$$

The discriminants of the two monic quadratics are $-4b^2$ and $-4d^2$.

Lemma 2.1 (Cross resultant). *The resultant of the two factors is*

$$\text{Res}(Q_+, Q_-) = 4[4u + (1 - u)^2 \eta^2]. \quad (5)$$

Proof. Evaluating Q_- at the two roots $a \pm ib$ of Q_+ gives

$$\text{Res}(Q_+, Q_-) = ((2a + ib)^2 + d^2)((2a - ib)^2 + d^2) \quad (6)$$

$$= (4a^2 - b^2 + d^2)^2 + 16a^2 b^2. \quad (7)$$

Using (2), the terms linear in η cancel and the expression reduces to (5). $\square$

3 Complete discriminant factorization

For monic polynomials,

$$\text{Disc}(Q_+ Q_-) = \text{Disc}(Q_+) \text{Disc}(Q_-) \text{Res}(Q_+, Q_-)^2. \quad (8)$$

Theorem 3.1 (Shape discriminant). *For the canonical quartet,*

$$\boxed{\text{Disc } Q_{u,\eta} = 256(1 - u)^2(1 - \eta^2)[4u + (1 - u)^2 \eta^2]^2.} \quad (9)$$

Proof. Since $b^2 d^2 = (1 - u)^2(1 - \eta^2)$, the product of the two quadratic discriminants is $16(1 - u)^2(1 - \eta^2)$. Multiplying by the square of (5) proves (9). $\square$

4 Degeneracy strata

All factors in (9) are nonnegative on the shape rectangle.

Corollary 4.1 (Complete zero set). *The quartet has a repeated root if and only if at least one of the following holds:*

1. $u = 1$;
2. $|\eta| = 1$;

3. $(u, \eta) = (0, 0)$.

Proof. The first two factors in (9) give the first two cases. The final square bracket vanishes only when both nonnegative terms $4u$ and $(1 - u)^2 \eta^2$ vanish, which is exactly $u = 0, \eta = 0$. $\square$

The geometry of each stratum is explicit:

$u = 1$ $a = 1$ and $b = d = 0$, giving roots $1, 1, -1, -1$.

$\eta = 1$ $d = 0$, so the negative-real pair coalesces at $-\sqrt{u}$; for $\eta = -1$, the positive-real pair coalesces at $+\sqrt{u}$.

$(u, \eta) = (0, 0)$

$a = 0$ and $b = d = 1$, so both quadratic factors equal $z^2 + 1$ and the roots $\pm i$ each have multiplicity two.

Intersections carry the corresponding higher overlap, but introduce no new zero set.

5 Uniform collapse along the axial boundary

Let $\mathcal{B} = \{1, 1, -1, -1\}$, with multiplicity, and let d_{match} be the minimum over root permutations of the maximum paired distance.

Theorem 5.1 (Uniform root-collapse estimate). *For every (u, η) in the shape rectangle,*

$$d_{\text{match}}(\text{roots } Q_{u, \eta}, \mathcal{B}) \leq \sqrt{2(1 - u)} + \frac{1 - u}{1 + \sqrt{u}}. \quad (10)$$

In particular the convergence to $\mathcal{B}$ is uniform in η as $u \rightarrow 1$.

Proof. Match the two roots with real part $a = \sqrt{u}$ to 1 and the two with real part $-a$ to -1 . For the positive pair,

$$|a \pm ib - 1| \leq |1 - a| + b \leq \frac{1 - u}{1 + \sqrt{u}} + \sqrt{2(1 - u)}, \quad (11)$$

because $b^2 \leq 2(1 - u)$. The same estimate holds for the negative pair using $d^2 \leq 2(1 - u)$. $\square$

The square-root term is natural at root level: imaginary heights vanish like $\sqrt{1 - u}$ in the worst transverse direction.

6 Coefficient collapse and equivalence

The boundary polynomial is

$$Q_{1, \eta}(z) = (z^2 - 1)^2 = z^4 - 2z^2 + 1, \quad (12)$$

independent of η .

Proposition 6.1 (Exact coefficient distance). *Let $C(u, \eta) = (1, 0, c_2, c_3, c_4)$ and $C_\star = (1, 0, -2, 0, 1)$. Then*

$$\|C(u, \eta) - C_\star\|_\infty = 4(1 - u). \quad (13)$$

Proof. The c_2 difference is exactly $4(1 - u)$. Moreover

$$|c_3| \leq 4\sqrt{u}(1 - u) \leq 4(1 - u), \text{ and } |c_4 - 1| \leq (1 - u)^2 \leq 4(1 - u). \quad (14)$$

Thus no other coefficient exceeds the c_2 difference. $\square$

Corollary 6.2 (Equivalent axial convergence). *For any sequence (u_n, η_n) in the shape rectangle, the following are equivalent:*

1. $u_n \rightarrow 1$;
2. $Q_{u_n, \eta_n} \rightarrow (z^2 - 1)^2$ coefficientwise;
3. the root multisets converge to $\mathcal{B}$ in matching distance.

Proof. Proposition 6.1 proves equivalence of the first two. Theorem 5.1 gives the first implies the third. Conversely, root convergence forces the positive real branch coordinate $\sqrt{u_n}$ to approach one. $\square$

7 Machine audit and physical position

Six hundred uniformly sampled parameter pairs verify Theorem 3.1 by direct root-product discriminants. The maximum relative error, normalized by $\max(1, |\text{Disc}|)$, is below 10^{-14} ; no violation of (10) occurs.

For the physical atlas, u rises from about 0.02 to about 0.718. Consequently the maximum-norm coefficient distance in (13) falls monotonically, but at the finest level it is still approximately 1.13. The matching distance to the double pair is about 0.577. These are directional observations, not evidence that the asymptotic regime has been reached.

8 Route consequence and claim boundary

The axial boundary is now mathematically understood. If a later analytic or validated argument proves $u_\sigma \rightarrow 1$, no separate control of η is needed for coefficient convergence: transverse dependence is uniformly suppressed. RH-217 quantifies that suppression differentially.

Even a proved fixed-quartet limit would still have only four roots, with two limiting locations. It cannot by itself provide a growing spectral divisor. No physical limit, determinant construction, Gate-A closure, Hilbert–Pólya operator, or zeta statement is asserted here.

References

- [1] Liang Wang. The centered conjugate-quartet shape manifold, 2026. RH-213 preprint.
- [2] Liang Wang. A finite monotone axial clock for the quartet shape flow, 2026. RH-214 preprint.
- [3] Liang Wang. Out-of-sample prediction of the finite shape clock, 2026. RH-215 preprint.